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CO₂-Based Fracturing Fluid: Evaluation of a Novel Foaming Agent Under High Temperature and Salinity Conditions

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Abstract

Hydraulic fracturing in unconventional reservoirs has traditionally relied on water-based fluids, but this approach suffers from excessive water consumption and environmental challenges. CO₂ foam fracturing fluids have emerged as a promising alternative, offering up to an 80% reduction in freshwater usage while improving proppant transport and formation cleanup. Therefore, this study develops a surfactant-stabilized CO₂ foam system using seawater as the base fluid, aiming to achieve high viscosity and stability of CO₂ under high-pressure, high-temperature (HPHT) conditions. The study evaluated foam rheology with a customized HPHT foam rheometer at 150 °C and 1,000 psi to assess the durability of foam's viscosity, which indicates its stability. Novel surfactant formulation was tested with and without the addition of a gel stabilizer or a high-temperature (HT) stabilizer. The CO₂ foam with the gel stabilizer achieved an initial viscosity of up to 55 cP at 100 /s shear rate, which was reduced by about 9% after 30 minutes and nearly 18% after 60 minutes at 150 °C. These results indicate significant improvement of the thermal stability compared to the surfactant-only foam (which dropped to 35 cP in 60 minutes). The optimized foam also demonstrated compatibility with high-salinity brine and maintained viscosity at shear rates up to 1000 /s, suggesting effective proppant suspension capability. By using low concentrations (0.5 and 1 wt%) of surfactant and readily available stabilizers, the formulation minimizes chemical loading. This work confirms the viability of a seawater-based CO₂ foam fracturing fluid that substantially reduces water usage while achieving the viscosity requirements for proppant transport in ultralow-permeability reservoirs. The findings provide a foundation for an environmentally sustainable and cost-effective foam fracturing technique to substitute the excessive use of fresh water in hydraulic fracturing.

Introduction

The increasing of global demand for energy has intensified the necessity for efficient and sustainable hydrocarbon extraction methodologies. While conventional water-based hydraulic fracturing has revolutionized the oil and gas industry, it has also encountered significant criticism due to its negative

environmental impact, particularly its high consumption of fresh water, the generation of flowback wastewater, and the potential contamination of groundwater (Arciniega-Esparza et al., 2022; Zhang et al., 2018). These challenges make the search for alternative fracturing techniques that minimize environmental risks while ensuring effective resource recovery a priority.

Among these alternatives, CO₂ foam fracturing has been presented as an attractive substitute for conventional water-based fracturing. It offers the potential to reduce environmental impact, particularly in water-scarce regions, during the extraction of unconventional hydrocarbons. (Abdelaal et al., 2024; Agarwal and Kudapa, 2023). CO₂ foam fracturing offers several benefits over conventional fracturing fluids, such as dramatically lower water consumption, better fracture extension, and enhanced hydrocarbon recovery (Lei et al., 2024; Samuel et al., 2023). The unique properties of CO₂ foam, such as low density and high diffusivity, help in developing complex fracture networks (Agarwal and Kudapa, 2023; Li et al., 2023). CO₂ foam fracturing also aligns with carbon management strategies because of the potential sequestration of CO₂ during operations (Sæle et al., 2022; Rossen et al., 2022). This technology shows improved sweep efficiency, greater residual trapping of CO₂, and less formation damage compared to water-based fluids (Zhao et al., 2023; Li et al., 2023).

High viscosity is important to control leak-off and ensure effective proppant transport, while CO₂ foam's compressibility and low water content may also help facilitate efficient flowback. The foam must be stable enough to carry proppants into the fracture network and prevent premature proppant settling. Premature proppant settlement results in screen-out, a condition in which proppants settle and block fluid flow, preventing further fracture extension and reducing hydrocarbon recovery.

Several factors affect CO₂ foam stability, including pressure, temperature, and surfactant concentration (Fu and Liu, 2021). Various approaches have been studied to enhance CO₂ foam stability, including polymer addition (Mingming and Shuzhong, 2015) and the use of nanoparticles to stabilize foam (Liang et al., 2018). Surfactant–nanoparticle systems have been found to improve foam stability by creating a robust interfacial film at the CO₂/water interface (Issakhov et al., 2022). A carefully optimized combination of sufficient foam viscosity, long-term stability, and proper foam mobility is necessary to achieve optimum fracture conductivity and overall productivity. Despite its advantages, CO₂ foam fracturing still faces challenges, such as the need to optimize surfactant performance, ensure long-term chemical stability of the foam, and demonstrate field-scale benefits (Rossen et al., 2022).

The primary objective is to achieve long-term foam stability and target viscosity for proppant transport at high temperatures. Specifically, maintaining performance for 1 hour under reservoir conditions, using minimal chemical additives. The 1-hour benchmark reflects typical operational timeframes for fluid residence in the wellbore and fracture during pumping stages and is commonly used in foam stability evaluations in fracturing studies (Al-Darweesh et al., 2022; Karadkar et al., 2023). In addition, the foam must be compatible with high-salinity brine and other common fracturing additives, allowing it to serve as a drop-in replacement for conventional fluids. Typically, it is common to develop a surfactant blend (including a self-foaming anionic surfactant) and incorporate low-cost stabilizers to enhance foam performance at 150 °C and high salinity. Then, the foam's rheology (viscosity vs. shear rate) is experimentally evaluated and foam dynamic stability, also known as the prolong test (foam half-life under shear) using state-of-the-art HPHT laboratory equipment. By combining insights from recent CO₂-foam advancements and our new experimental results, this study demonstrates a path toward a more sustainable fracturing fluid and identifies remaining gaps for future improvement. The following sections describe the materials and methodology, present the results and discussion of the foam performance, and finally conclude the feasibility of this advanced CO₂ foam system for hydraulic fracturing applications.

Materials and Methodology

Materials

In this study, a self-foaming polymer (SFP) was utilized to generate stable and high foam viscosity at high temperature. It is made of alkyl alcohol sulfate. This SFP surfactant was selected for its cost-effectiveness and its ability to generate stable foam in high-salinity CO₂ systems. A 10 gpt of gel stabilizer and a 10 gpt of high temperature stabilizer (sodium thiosulfate) were used. A synthetic seawater brine was prepared as the base liquid to simulate field conditions and eliminate freshwater usage. The seawater composition is detailed in [Table 1](#), matching the chemical composition of typical Gulf seawater. Using a high-salinity brine allowed us to assess foam stability under worst-case water chemistry conditions and ensured the foam's compatibility with seawater as a fracturing fluid.

Table 1—Sea water composition used in this study.

Ions	Seawater (ppm)
Sodium	17,490
Chloride	32,710
Calcium	600
Magnesium	2,120
Bicarbonate	150
Sulfate	4,700
Potassium	780
TDS	58,550

Four formulations were tested in this study, as illustrated in [Table 2](#):

Table 2—Foam Formulations and Additive Composition Used in This Study

Test No.	SFP Concentration	Gel Stabilizer	HT Stabilizer	Notes
1	1.0 wt%	None	None	Higher surfactant concentration to assess the viscosity increase
2	0.5 wt%	None	None	Baseline (Control) formulation
3	0.5 wt%	10 gpt	None	With gel stabilizer
4	0.5 wt%	None	10 gpt	With sodium thiosulfate as a high temperature stabilizer

The 0.5 wt% and 1.0 wt% SFP concentrations were selected from an initial screening (0.1–2 wt%) to balance foam performance with chemical economy.

Gel Stabilizer: 0.5 wt% SFP + 10 gpt (gallons per thousand gallons) of a gel stabilizer additive, which in this case was sodium thiosulfate. In our formulation, adding 10 gpt thiosulfate was hypothesized to improve foam stability in high salinity and at elevated temperature by maintaining surfactant efficacy.

High-Temperature Stabilizer: 0.5 wt% SFP + 10 gpt of a high-temperature (HT) stabilizer additive. This proprietary HT stabilizer is designed to enhance foam longevity at elevated temperatures (exceeding 100 °C).

Experimental Apparatus

Foam generation and testing were carried out in a customed high-pressure, high-temperature (HPHT) foam rheometer. The apparatus is capable of operating up to 6,000 psi and 177 °C. It includes dual cylinders (Quizix pumps) for precise injection of liquid and CO₂, a high-pressure foam generator/mixer, a recirculating loop with flow and pressure sensors, and a viewing cell for observing foam texture. Fig 1 provides a schematic of this HPHT foam rheometer. In a typical test, the surfactant solution (with or without stabilizers) was loaded into the liquid accumulator while CO₂ was pressurized in a gas cylinder. The fluids were co-injected at a controlled ratio to achieve the desired foam quality (in these experiments, fg 0.70, or 70% CO₂ by volume). The mixing of CO₂ and surfactant brine through a high-shear foam generator produced a fine-textured foam, which was circulated continuously in the loop. Differential pressure across a pipe section in the loop was measured to determine the apparent foam viscosity via the Hagen–Poiseuille relationship (corrected for two-phase flow). The rheometer allowed imposition of various shear rates by adjusting the flow rate; the foam rheology was examined from 100 /s up to 1000 /s to cover the range from moderate shearing (in wide fractures) to high shearing (through perforation tunnels). The entire loop was enclosed in a heating chamber, which was set to the target temperature of 150 °C for these tests.

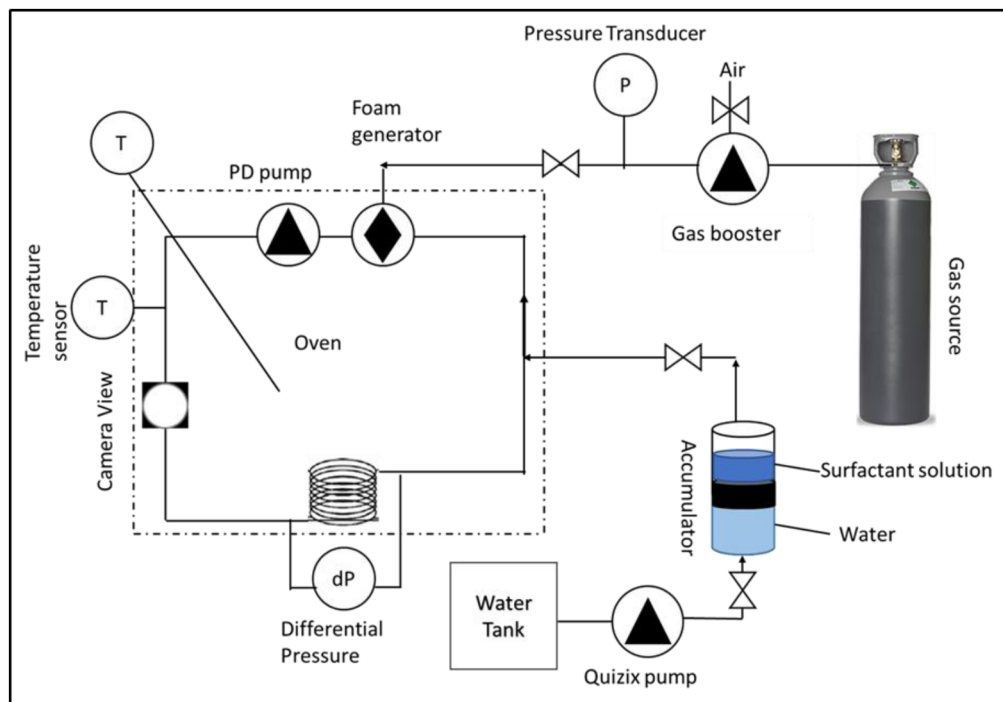


Figure 1—A schematic of the HPHT Foam rheometer (Al-Darweesh et al., 2022).

Test Procedure

Once the foam was generated and the system reached thermal equilibrium, we first measured the foam's steady-shear viscosity at several shear rates (100, 200, 400 /s, etc.) under constant conditions (150°C, 1000 psi, 70% quality). This provided insights into the foam's pseudoplastic (shear-thinning) behavior and baseline viscosity level for each formulation. Next, we evaluated foam stability over time by holding the foam at a constant shear rate (100 /s) and recording the viscosity decay for at least 60 minutes. This dynamic stability test at 150 °C and 1000 psi simulates the requirement for the foam to sustain proppant in suspension during a fracturing treatment (which typically lasts on the order of an hour or more). No proppant was added in these preliminary tests (foam was tested without suspended sand); however, the high viscoelasticity of the foam at low shear effectively traps some micro-bubbles in the loop that serve as visual tracers of stability.

Results and Discussion

Foam Rheology at 150 °C (Initial Viscosity)

We first compare the initial viscosity of the CO₂ foam for the four formulations (baseline vs. +stabilizers) under HPHT conditions. All measurements here were conducted at 150 °C, 1000 psi, and 70% foam quality.

Higher surfactant concentration of 1wt% SFP:

At a higher surfactant loading, the 1 wt% SFP formulation achieved a significantly high apparent viscosity of around 138 cP at 100 /s—more than three times that of the 0.5 wt% case, as illustrated in Fig 2. The substantial increase in viscosity observed for the 1 wt% SFP foam is primarily attributed to enhanced lamella strength and more robust interfacial films resulting from a higher micelle density. At this concentration, the surfactant is likely near or above its critical micelle concentration (CMC), which promotes the formation of micelles that contribute to stronger foam structures and more stable films. In addition to thicker liquid films, the elevated surfactant concentration improves the foam's ability to resist shear deformation through dynamic film repair mechanisms. Under shear, deformation at the bubble interface generates surface tension gradients, which in turn drive surfactant micelles from regions of high concentration to low concentration, effectively restoring film uniformity and stabilizing the structure. This mechanism, combined with increased lamella viscosity and finer bubble texture, contributes to the foam's shear-thinning behavior, with viscosity decreasing gradually to approximately 49 cP at 1000 /s. These rheological characteristics suggest strong structural integrity under flow, which is favorable for proppant transport. However, despite these benefits, high surfactant loadings can pose field-scale challenges, such as increased cost, potential formation damage, and compatibility issues under high-salinity or high-temperature reservoir conditions.

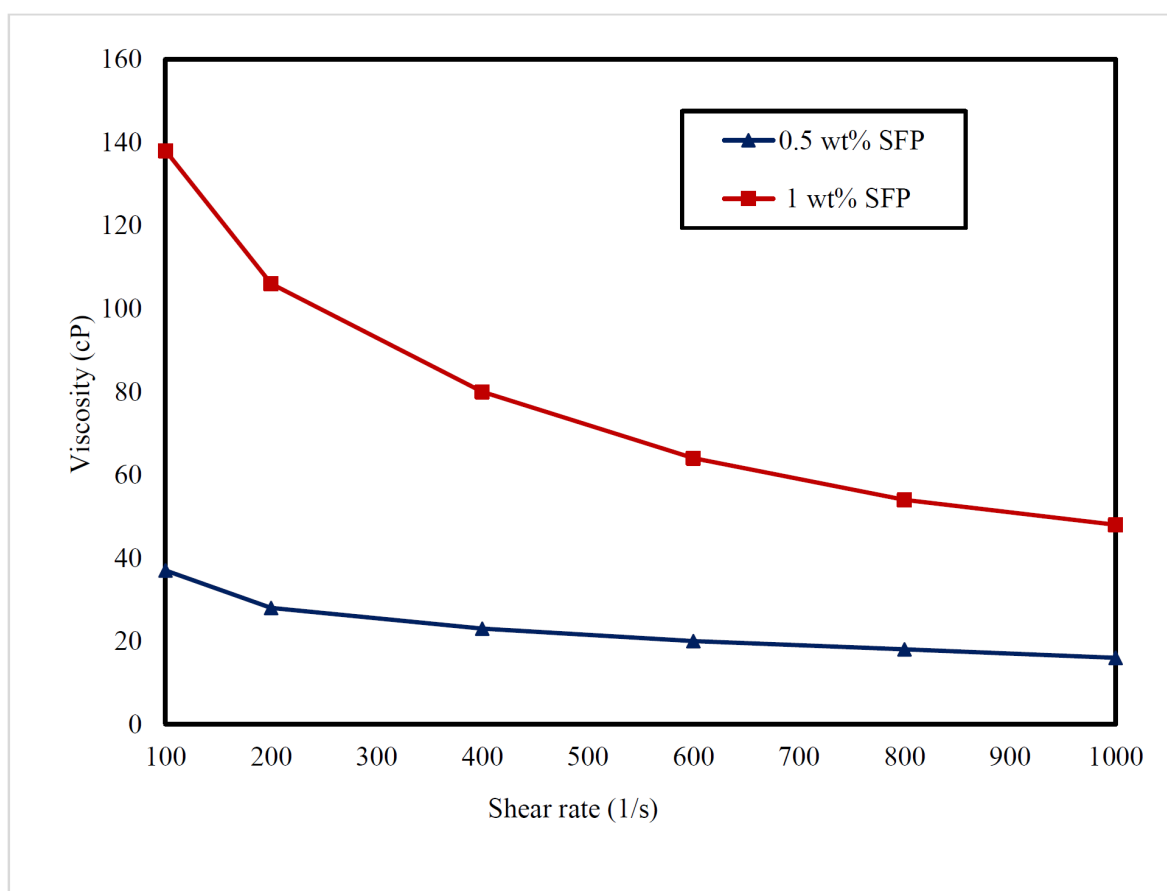


Figure 2—Foam viscosity with 0.5wt% and 1wt% SFP at 150 °C, 1000 psi, and 70% foam quality.

Baseline 0.5% SFP:

To minimize chemical usage and reduce these potential drawbacks, the remainder of this study focuses on 0.5 wt% SFP as the base formulation. As shown in Fig 2, this baseline foam exhibits a viscosity of 38 cP at 100 /s, with values falling below 20 cP at 600/s. The pronounced shear-thinning behavior reflects typical characteristics of foams and non-Newtonian fluids, where foam films thin and align under stress, reducing resistance to flow. While this base case provides a useful benchmark, its low viscosity is inadequate for effective proppant transport on its own.

0.5% SFP with the gel stabilizers (Fig 3):

0.5 wt% SFP + 10 gpt gel stabilizer ($\text{Na}_2\text{S}_2\text{O}_3$) improved low-shear viscosity to 50 cP. This enhancement is because gel stabilizer is used to prevent the oxidation of polymer which breaks the polymer chain and reduces the viscosity. 0.5 wt% SFP + 10 gpt HT stabilizer yielded the highest viscosity of 55 cP at 100 /s. The HT additive, designed for thermal stability, likely forms resilient micellar or colloidal structures that reinforce foam films. Viscosity at 600 /s remained around 30 cP, with indications of yield-stress behavior at low shear.

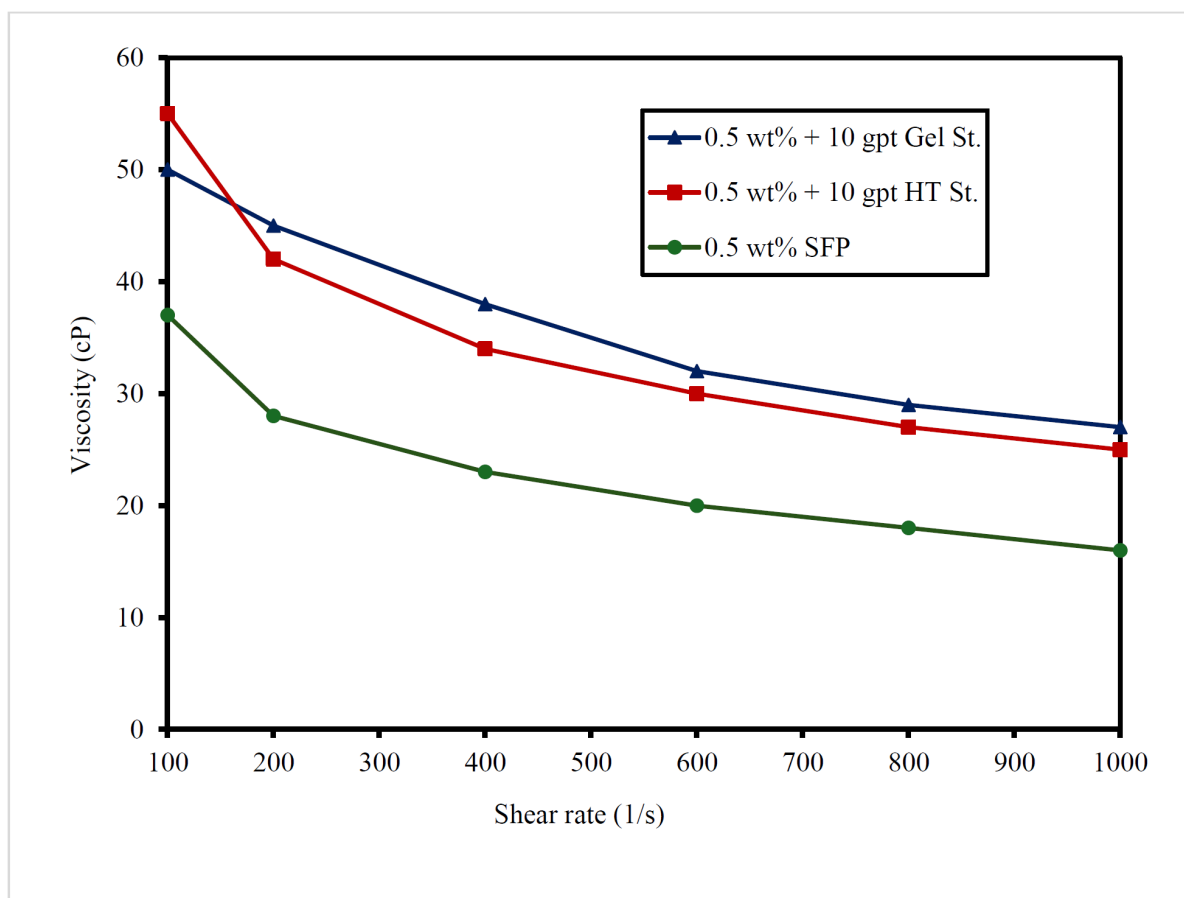


Figure 3—Foam viscosity at 150 °C, 1000 psi, and 70% foam quality.

Mechanistic Insight: The enhancement in foam viscosity with the stabilizers can be attributed to their combined effects on foam structure and stability:

- **Interfacial film reinforcement:** Both additives likely contribute to strengthening the gas–liquid interface. Thiosulfate may improve film stability indirectly by interacting with divalent ions (e.g., Ca^{2+} , Mg^{2+}) that would otherwise disrupt the surfactant film. The high-temperature (HT) stabilizer, on the other hand, may adsorb directly at the interface, forming more robust interfacial layers. These

reinforced films help suppress gas diffusion and reduce Ostwald ripening, where larger bubbles grow at the expense of smaller ones.

Thermal protection: The HT additive appears to offer protection against surfactant degradation at elevated temperatures (up to 150 °C). This could occur through mechanisms such as modification of the critical micelle concentration (CMC) or formation of thermally stable colloidal aggregates. These structures help preserve foam integrity even under conditions where conventional surfactants would begin to break down.

These mechanisms are consistent with the elevated initial viscosities observed in stabilized formulations. While the foam viscosity achieved with stabilizers (50–55 cP) remains below the ideal 100+ cP range typical for conventional fracturing fluids, it represents a meaningful improvement over the 38 cP baseline. In practical terms, a foam with 55 cP could be suitable for transporting ultra-lightweight proppants (e.g., porous ceramic beads) in moderate-width fractures. However, it may be insufficient for carrying standard sand across large or high-flow fractures—highlighting the need for further optimization or a return to higher surfactant concentrations in those scenarios.

Foam Stability at 150 °C (Thermal Stability over Time)

A critical requirement for fracturing fluids is to maintain viscosity over the pumping and placement timeframe (on the order of an hour or more). We monitored the foam viscosity as a function of time at 150 °C for 60 minutes (at constant 100 /s shear) to evaluate the thermal stability of each formulation. The results are summarized as follows (viscosities at $t=0$ and after 60 min):

High surfactant concentration- 1wt% SFP: At this elevated concentration, the foam started at 140 cP and showed a modest decline to 129 cP over 60 minutes a loss of only 8%, as shown in Fig 4. The curve reflects a relatively stable foam structure, with viscosity plateauing for several intervals before gradually declining. This suggests that at high surfactant loading, the foam films are more robust and resistant to thermal thinning and gas diffusion. However, while this performance is promising, such high concentrations are not always practical due to cost, environmental concerns, and potential compatibility issues, highlighting the need to assess performance at lower surfactant concentration levels.

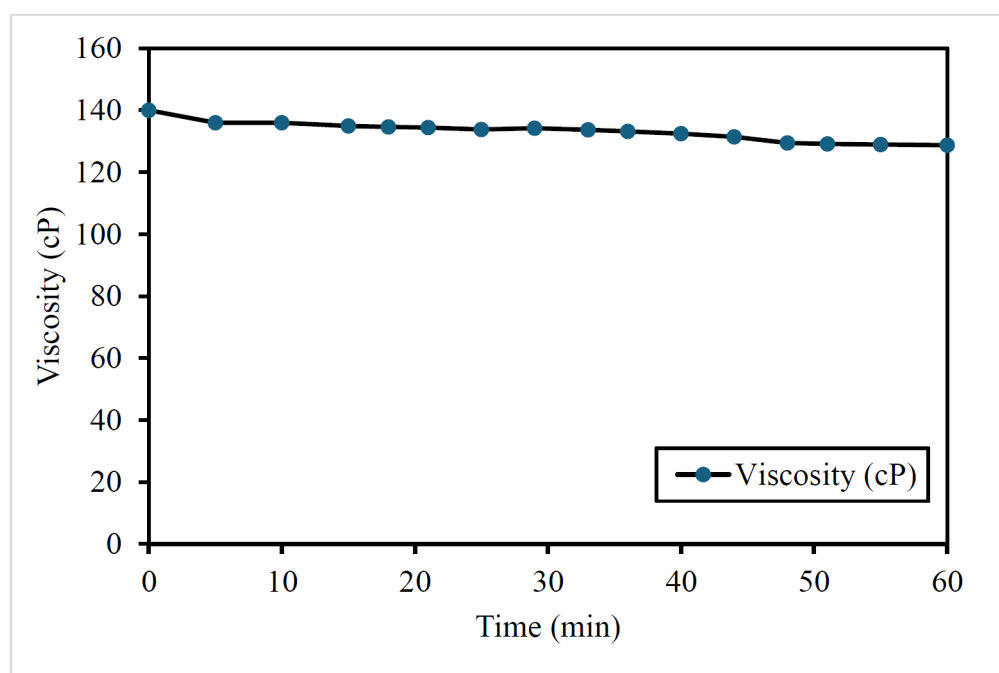


Figure 4—Thermal stability of Foam viscosity with 1wt% at 150 °C, 1000 psi , 100/s shear rate, and 70% foam quality.

Baseline 0.5 wt% SFP: The initial viscosity was 45 cP—slightly higher than observed in the shear tests (38 cP), possibly due to foam regeneration in the circulation loop. However, this viscosity steadily declined to 35 cP over 60 minutes, representing a 22% loss. This continuous drop is because micelles thermally decomposed due to the high temperature. Mechanistically, thermal energy likely disrupts surfactant adsorption at the gas–liquid interface and promotes CO₂ dissolution into the aqueous phase. At 60 minutes, the foam showed visible coarsening in the sight cell and minimal resistance to flow, suggesting poor long-term viability for proppant suspension at this temperature.

Gel Stabilizer (0.5 wt% SFP + 10 gpt Na₂S₂O₃): The foam began at 55 cP, dropped to 45 cP within the first 10 minutes, then decreases to around 40 cP for the remaining hour. The stabilizer's helps prevent destabilizing precipitation, while its weak gelation likely retains liquid within the foam films. Although the foam retained 73% of its initial viscosity, its absolute viscosity remained higher than the baseline throughout the test, offering better—but still limited—thermal endurance.

High-Temperature Stabilizer (0.5 wt% SFP + 10 gpt HT additive): This formulation showed the best overall performance, starting at 55 cP and maintaining 50 cP for the first 30 minutes. At 60 minutes, viscosity declined only slightly to 45 cP, retaining 90% of the initial value. This plateau-like behavior suggests that the HT additive effectively resists the typical thermal degradation processes observed in surfactant foams. Stabilization mechanisms include (1) adsorption at the interface to form resilient barrier films, and (2) stabilization of micellar structures, which protects the surfactant from thermal breakdown or prevents its desorption. Visual observation also confirmed that the HT-stabilized foam maintained a fine bubble structure and no significant liquid phase separation by the end of the test.

To visualize the stability trends, Fig 5 plotted viscosity vs. time for the three cases. The baseline line dropped downward continuously, whereas the stabilized foams plateaued at higher viscosities. The HT stabilizer curve was almost flat for 30 min, then only gently dropped. These results underscore the value of chemical stabilizers: the HT additive in particular provided thermal endurance that pure surfactant lacked.

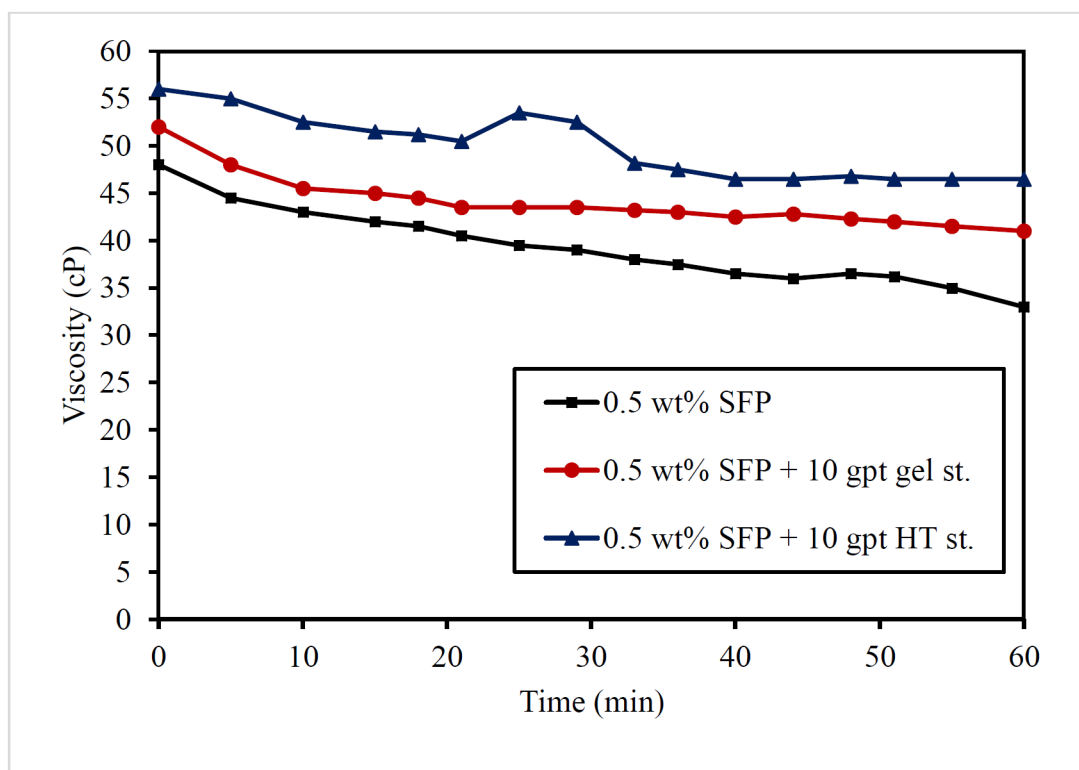


Figure 5—Thermal stability of Foam viscosity with 0.5 wt% at 150 °C, 1000 psi , 100/s shear rate, and 70% foam quality.

Implications for Fracturing Performance

The above rheology and stability results have several practical implications for using CO₂ foam in hydraulic fracturing:

Proppant Transport Efficiency: Achieving sufficient viscosity is critical for suspending proppant in the fracture (Che et al., 2025; Wang and Elsworth, 2020). Our HT-stabilized foam reached 55 cP at the start of injection and was still 45 cP after an hour. This viscosity is likely adequate to carry lightweight proppants in shallow or moderate-depth fractures. For standard sand (which is heavier), 45 cP may be on the lower end; field practices often target 100+ cP with crosslinked gels. However, the foam's non-Newtonian characteristics (yield stress and elasticity) mean that even at tens of cP it can hold particles in suspension. (Gao et al., 2024; Wang and Elsworth, 2020).

Additionally, foam has a high apparent viscosity at low shear – in stagnant conditions it behaves like a gel (Gao et al., 2024). The shear-thinning behavior observed (viscosity dropping to 30–35 cP at high shear) is beneficial for pumping due to reduced friction in surface equipment and tubing. Our results indicate that further increasing the surfactant concentration or optimizing the stabilizer dosage could push the foam's viscosity closer to the desired range for heavier proppants. Nonetheless, even in its current form, the foam could transport proppant in certain scenarios, and the absence of significant settling (no "screen-out" observed in our loop tests) is encouraging.

Seawater compatibility: The ability of the foam to perform well in high-salinity synthetic seawater (TDS 58,550 ppm) is a key outcome of this study. Initial screening demonstrated that the SFP-based foam could be generated and remained stable without collapse, even in such challenging brine conditions. While high salinity is known to destabilize many conventional foams, this is not entirely true for certain surfactants. In our case, the SFP surfactant showed favorable compatibility with seawater, and the addition of sodium thiosulfate played a critical role. By chelating divalent ions like Ca²⁺ and Mg²⁺, thiosulfate likely prevented surfactant precipitation and helped maintain foaminess. Notably, similar foam performance with viscoelastic surfactants (VES) has required concentrations as high as 6 wt% (Al-Darweesh et al., 2024), whereas our system achieved equivalent or superior performance at just 1 wt% SFP. These results confirm that freshwater can be fully replaced with seawater as the base fluid, achieving the environmental goal of reducing freshwater usage. This also opens opportunities for field applications using seawater or even produced water (TDS 24,611 ppm), reducing the need for costly freshwater transport. Our findings support the feasibility of seawater-based fracturing fluids and demonstrate an effective strategy to overcome the water-compatibility challenge in foam design.

Economic and environmental feasibility: The formulated foam system uses relatively low additive concentrations, which is advantageous for both cost and environmental footprint. The surfactant concentration is 0.5 wt%, and each stabilizer is used at 10 gpt (approximately 1% by volume). In comparison, other approaches to stabilize foams (e.g., nanoparticle + polymer systems) might require >1 wt% nanoparticles plus hundreds of ppm of polymers. Such high-load systems raise concerns about expense and potential formation damage (e.g., plugging pores with residual particles or gel) (Mansha et al., 2023). In contrast, our SFP-based foam stays stable even with much less solid content. This should translate to lower chemical costs per job. Furthermore, by cutting 80% of the water volume, logistics and water-handling costs are greatly reduced—fewer water tanker trucks, less flowback water for disposal, etc. Environmentally, using CO₂ (which can be captured for sequestration) and saving water are clear wins. If captured industrial CO₂ is used, the net carbon footprint of the fracturing operation could be lower than that of a conventional water-based frac (which indirectly involves CO₂ emissions through water treatment, hauling, etc.). Of course, CO₂ supply has its own costs and logistics, but regions with available CO₂ (e.g., from nearby gas processing or fermentation plants) could find this trade-off attractive. In summary, our preliminary results validate that a seawater-based, low-chemical CO₂ foam is viable, addressing both water scarcity and chemical usage concerns. More optimization is needed before field deployment—for instance,

identifying the optimal surfactant blend or incorporating a small amount of polymer for even longer stability—but the concept appears to be on firm footing.

Conclusions

This study demonstrated the technical viability and environmental advantages of a CO₂-based fracturing fluid formulated with seawater and stabilized by a novel surfactant system. The optimized foam, composed of 0.5 wt% self-foaming surfactant and either a gel or high-temperature stabilizer, achieved viscosities up to 55 cP at 150 °C and 1000 psi, significantly higher than the baseline surfactant-only formulation. Increasing the surfactant concentration to 1 wt% further enhanced viscosity, reaching up to 135 cP under the same conditions. Such a concentration remains within a reasonable range for hydraulic fracturing operations, offering a practical pathway to tailor fluid properties based on reservoir demands. The foam exhibited favorable shear-thinning behavior for effective pumping and maintained up to 90% of its initial viscosity after 60 minutes, indicating strong thermal stability. Compatibility with high-salinity seawater was also confirmed, reinforcing its potential for use in water-limited environments. These findings point to a more sustainable fracturing fluid system that significantly reduces freshwater use, lowers chemical loadings, and offers potential means for CO₂ sequestration.

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